An

Internship Report

by

Prajwal Kulkarni

Neuro-Engineering Lab

Neurosurgery Education and Training School

Department of Neurosurgery

All India Institute of Medical Science, New Delhi

May 2025 - September 2025

DECLARATION

I, "**Prajwal Kulkarni**" hereby declare that the internship report for the internship at **Neuro-Engineering Lab, Neurosurgery Education and Training School, Department of Neurosurgery, All India Institute of Medical Sciences** is an original record of the work carried out by me under the supervision of **Dr.Rohan Dhanakshirur** from **May 2025 – September 2025.** The matter embodied in this report has not been submitted elsewhere for the award of any degree or diploma.

Prajwal Anand Kulkarni
Research Intern
Neuro Engineering Lab,
NETS, Department of Neurosurgery,
All India Institute of Medical Sciences, New Delhi

PREFACE

With the current times where AI is the new norm in every industry, AI also finds its application in a very sophisticated domain such as neurosurgery.In our case it is particularly in the “Evaluation of Neurosurgical Training” where doctors undergo simulated surgical training for cylindrical suturing (i.e. suturing performed on synthetic tubes , nerves and vessels) and craniotomy (Drilling performed on synthetic material, sheep head, scapula)

This report covers the work done on a project in which neurosurgical training samples are evaluated and scored by a machine learning model. My work revolved around deployment of the model through a stand-alone application which not only provided a user interface but also a complete pipeline including features like report generation, grad-cam etc.

Another project which we worked on is the project on Evaluation of Cylindrical Sutures, Here the approach involved using agentic AI to provide evaluation for the cylindrical suturing training samples, with an interesting feature of descriptive report generation.

Lastly the report also contains the work on Sensor Fusion for Neurosurgical Endoscopic Experiment in which I particularly dealt with eye-tracking of neurosurgeons.This experiment aims to study the patterns in the eye movement of neurosurgeons Additionally It also aims to study the difference in movement paths of eyes across neurosurgeons with range of experience.

ACKNOWLEDGEMENT

I would like to take this opportunity to express my sincere gratitude to those who have been an integral part of this journey. While it is not possible to acknowledge everyone in this brief note, I will make an earnest attempt.

First and foremost, I extend my heartfelt appreciation to **Prof.(Dr.) Ashish Suri**, Head of Unit 2, and **Dr. Ramandeep Singh**, Senior Scientist, for providing me the opportunity to pursue my internship at the Department of Neurosurgery, AIIMS. Their support has been invaluable.

I am especially grateful to **Dr. Rohan Dhanakshirur**, Ph.D., IIT Delhi. My supervisor during the internship, for his guidance and patience. His willingness to overlook my mistakes and provide assistance at various stages greatly contributed to my learning experience.

I would also like to acknowledge **Mr.Anuj Saini**, whose mentorship and insights were instrumental in shaping my understanding of key aspects of the work.

I would like to thank **Dr. Pavan Kunchur**, SPOC from KLS Gogte Institute of Technology, for facilitating this opportunity and ensuring a seamless coordination process.

Finally I would like to thank all the other technical and non-technical members of the lab for their support.

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Organization profile

All India Institute of Medical Sciences, New Delhi

All India Institute of Medical Sciences was established as an institution of national importance by an Act of Parliament with the objects to develop patterns of teaching in Undergraduate and Post-graduate Medical Education in all its branches so as to demonstrate a high standard of Medical Education in India; to bring together in one place educational facilities of the highest order for the training of personnel in all important branches of health activity; and to attain self-sufficiency in Post-graduate Medical Education.

Objectives of AIIMS

- To develop a pattern of teaching in undergraduate and postgraduate medical education in all its branches so as to demonstrate a high standard of medical education to all medical colleges and other allied institutions in India.
- To bring together in one place educational facilities of the highest order for the training of the personnel in all important branches of the health activity.
- To attain self-sufficiency in postgraduate medical education.

Neurosurgery Education and Training School

Neurosurgery Education and Training School is an initiative taken by the Department of Neurosurgery of the All India Institute of Medical Sciences, New Delhi to disseminate information and supplement the surgical knowledge of the trainee and trained neurosurgeons all over the globe. It also intends to network the neurosurgeons working in different parts of the world through a discussion forum.

The virtual education material on the website of NETS, in the form of video library, webinars and tele-education material shall update the neurosurgeons as to the recent advances in the relevant subject. Also, the interaction of the inept trainee neurosurgeons with the experienced and masters in the field shall provide an enriched mingling of anatomy, pharmacology, pathology, radiology, neurology and surgical techniques related to neurosurgery.

NETS has a well-equipped Neurosurgery Skills Training Facility where regular skills training sessions, short-term skills training programs and quarterly Indo-German workshops are conducted to provide more detailed hands-on skills training to trainee MCh, DNB and MD, as well as certified neurosurgery professionals coming from different parts of India and other countries.

Cylindrical Suturing Evaluation

Neurosurgical training starts with suturing on flat sheets where the main task is to suture a cut made on the synthetic sheet and after 10 iterations of this training by a doctor with increasingly thinner threads as they move up the iteration count ,The doctor can perform the training on a synthetic tube and then proceeded with live rats and chicken wings for nerve and vessel suturing.

Problem Statement

The neurosurgical training done by doctors in the NETS lab requires scoring and feedback for understanding the progress and also to understand the error in the course of action and to realise the room for improvement in specific areas like position, angulation etc.

The task at hand was to provide such an evaluation for cylindrical i.e. synthetic tube, nerve , vessel suturing training samples and also to provide features like report generation and video recording through a stand-alone application.

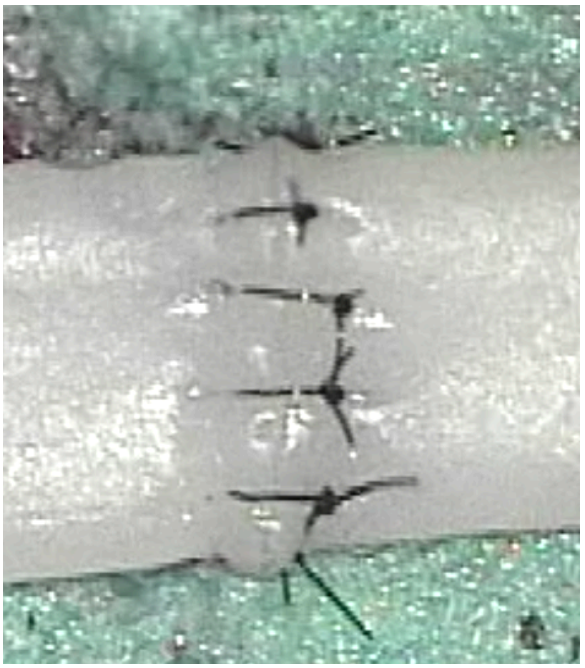
Methodology

Data

AI is a data problem and for this cylindrical evaluation problem I had to collect data from the neurosurgical training videos through the NETS server. All the training recorded videos were watched from 2015 till present (20th May, 2025) and images were obtained by capturing the screen after the final suture. In cylindrical suturing the doctor performs suturing on one side and then flips the cylindrical tube and sutures the other side. This process ensures complete suture at all areas throughout the cylindrical joining and due to this 2 images per suturing training iteration are obtained.

The data is collected from training videos of short term trainees and residents from neurosurgery and other departments as well. In total the obtained data can be classified into categories like nerves ,vessels, synthetic tubes normal and synthetic tubes bypass .

The data consists of a total 514 images which includes 394 joining (synthetic tubes), 26 images of bypass (synthetic tubes) and 94 images of nerves and vessels combined.



(a)

(b)

(c)

(a) : Joining suture sample - Dr.Anil Jadhav (Trainee 2016)

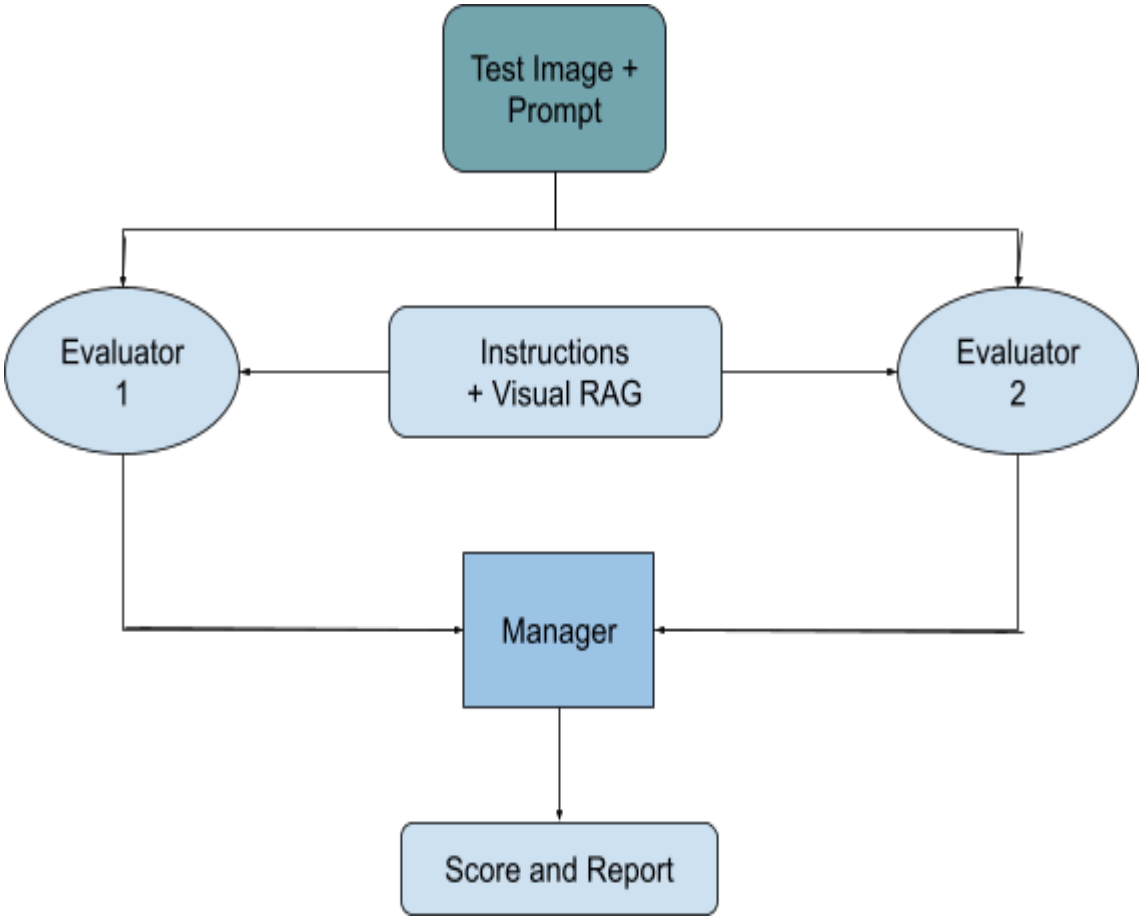
(b) : Bypass suture sample - Dr.Raghav Singhla (Resident)

(c) : Nerve/Vessel suture sample - Dr.Bhaskar (Resident)

Agentic AI

After obtaining the data, We decided to use agentic AI for the evaluation of these suture samples, agents refer to instances of large language models i.e. Its core functionality and responses are derived from the LLM but with that they additionally have decision making capabilities which makes them more effective for automating tasks.

We used multiple agents for the task as the task involved evaluation as well as an analysis of evaluation, by analysis we mean a detailed report explaining all the scores provided by the agent and also giving recommendations on what can be improved.



(d)

As the above flowchart describes the control flow and process of evaluation of images through multiple agents, we shall try to explain multi agent and manager architecture and also touch upon the idea of visual RAG (Retrieval Augmented Generation).

Manager - Agent Architecture

At its core, the architecture consists of two primary roles:

The Manager (or Orchestrator) Agent**:** This is the central coordinator which handles all the experts or agents, in our case we had 2 experts reporting to the manager who would finally give out the output as per the instructions.

The Worker (or Expert) Agents: These are specialized agents, who are allocated work accordingly like in our case to provide evaluation and report for a test image.

We had used two open source multimodal large language models for our manager and expert architecture from llama, llama scout and llama maverick where the maverick was used for manager role and an expert role where scout was used as the other expert. Here llama maverick was used for manager role because it is a comparatively larger model with better multimodal benchmarks. Considering the GPU constraint and cost of a paid model per request, we chose to use these models for evaluation using API calls from Groq and used Agno framework to handle images as inputs.

Visual Retrieval Augmented Generation

Visual RAG is still the holy grail in LLM research where there's no go to method for small vision or multimodal models to be provided with image data and annotations such that If I provide a pdf with images and annotations the model cannot refer to the 3 closest images with respect to suture quality and then give precise evaluation based on RAG.

Hence we used the CLIP vision transformer model to convert the document into vector embedding and then provide that to the system as a knowledge base where we had 10 images in the beginning and later 50 images were converted and provided to the agents.

According to the instructions provided to the agents, After getting the inference image, they would search the knowledge base for three most similar images and then consider them as gold standard but not copy the scores or provide their average instead give their own evaluation and interpretation based on the knowledge base.

Results and Conclusion

We analysed the scores provided by the agentic system for 50 test images and the expert scores obtained from the neurosurgeon as gold standard. The analysis outcomes are given below :

- Root mean square error between agentic system final score and expert final score = 2.2 out of 10
- Root mean square error between expert final score and average of subscores by expert = 2.5833 out of 10
- Plus/minus 1 Accuracy = 52%

Here Plus/minus 1 Accuracy refers to the Percentage of the predicted scores by the model being in +1 or -1 range of scores given by expert or surgeon.

The obtained results are underwhelming and can be improved significantly.One of the reasons for taking up this approach was to explore ways in which this evaluation project can be done without annotations and LLMs which are pretrained on general data could possess capabilities to do such a task but the results obtained indicate it's difficult to expect small multimodal models to perform good enough for such an sophisticated and niche problem.

Hence we intend to continue working on this project and explore more approaches to solve the problem.

CRANIOTOMY EVALUATION MODEL DEPLOYMENT

Craniotomy refers to a surgical procedure where a surgeon temporarily removes a section of the skull to access and operate on the brain. It is a prerequisite for a neurosurgeon as, In almost all neurosurgeries craniotomy has to be performed and for such a high risk task proper training and practice are mandatory with active feedback.

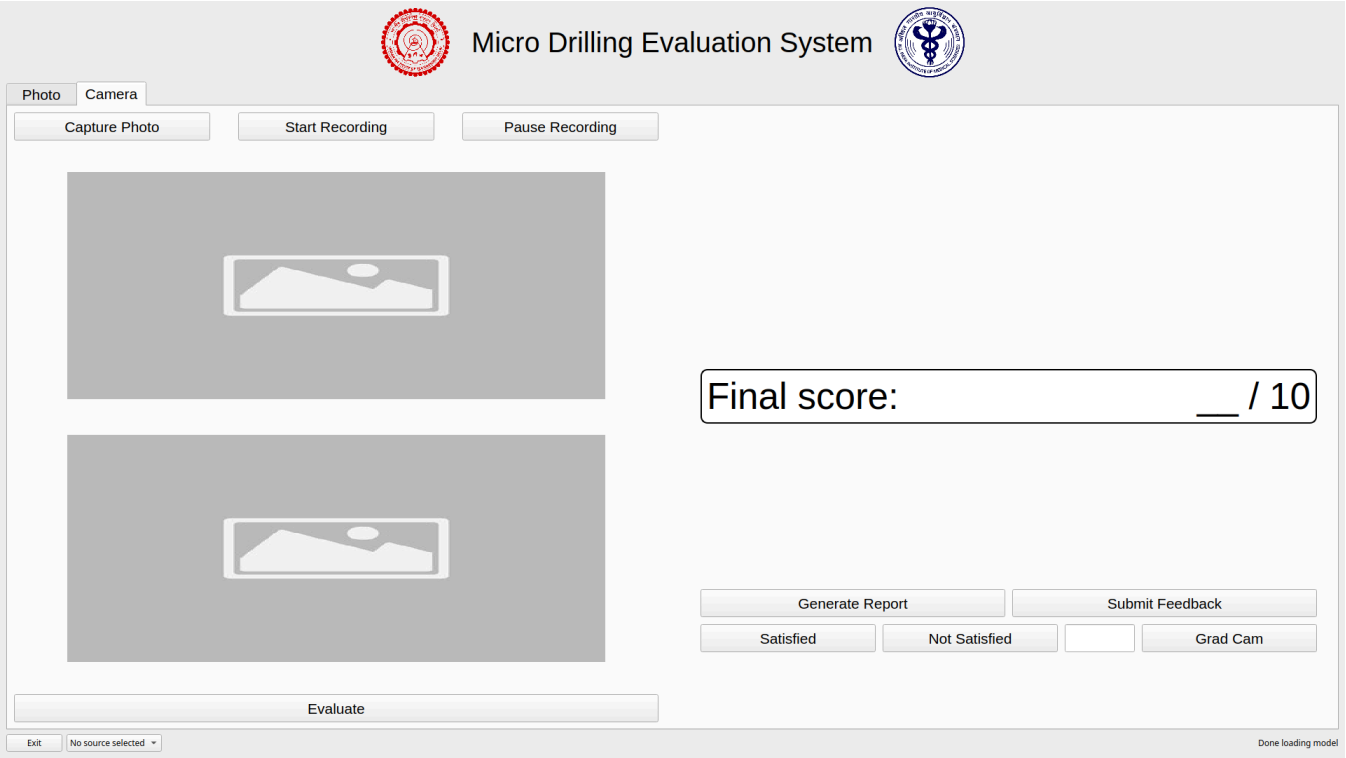
Problem statement

The key task in this project was to develop and deploy a stand-alone application which could provide real time evaluation for the training samples of doctors with video recording, report generation, feedback and grad-cam visualization like features implemented.

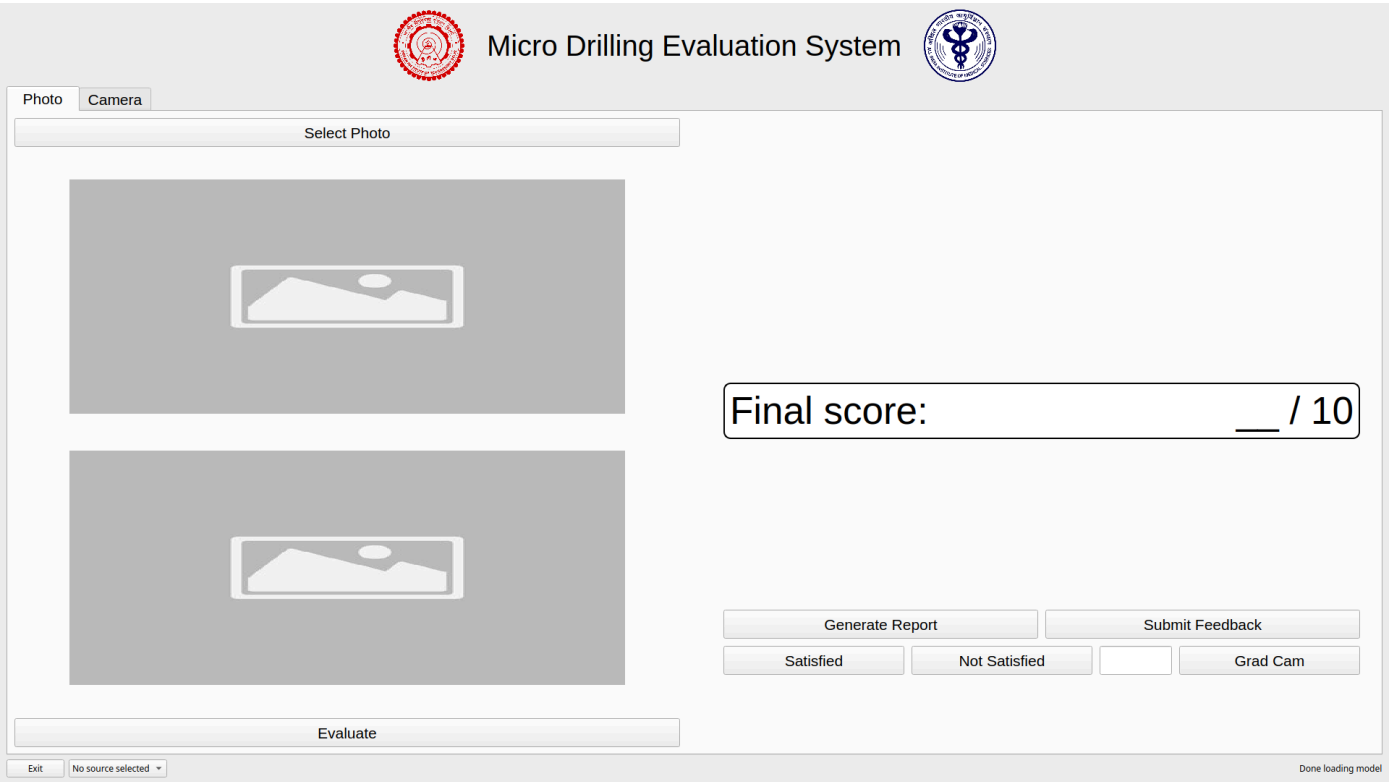
Methodology

Python is the language used along with HTML and CSS for the format of the report and PyQt5 is the framework used in this project for building the front-end which is convenient to handle and work with python libraries like opencv to infer the model and also as it has signal slot architecture and straight forward widget layout format. We have made sure that the front end feels intuitive and easy to use for everyone.

The User Interface has two different tabs: photo mode and camera mode. In photo mode the user or doctors can upload a static image which is already saved in the device and in the camera mode live feed recording can be done and options to change the camera source are also provided. The camera feed is handled through a thread which is created every time a user starts the camera or uses camera mode.Additionally the live feed of the selected camera can be captured and then evaluated in camera mode.



(f) Camera tab view



(g) Photo tab view

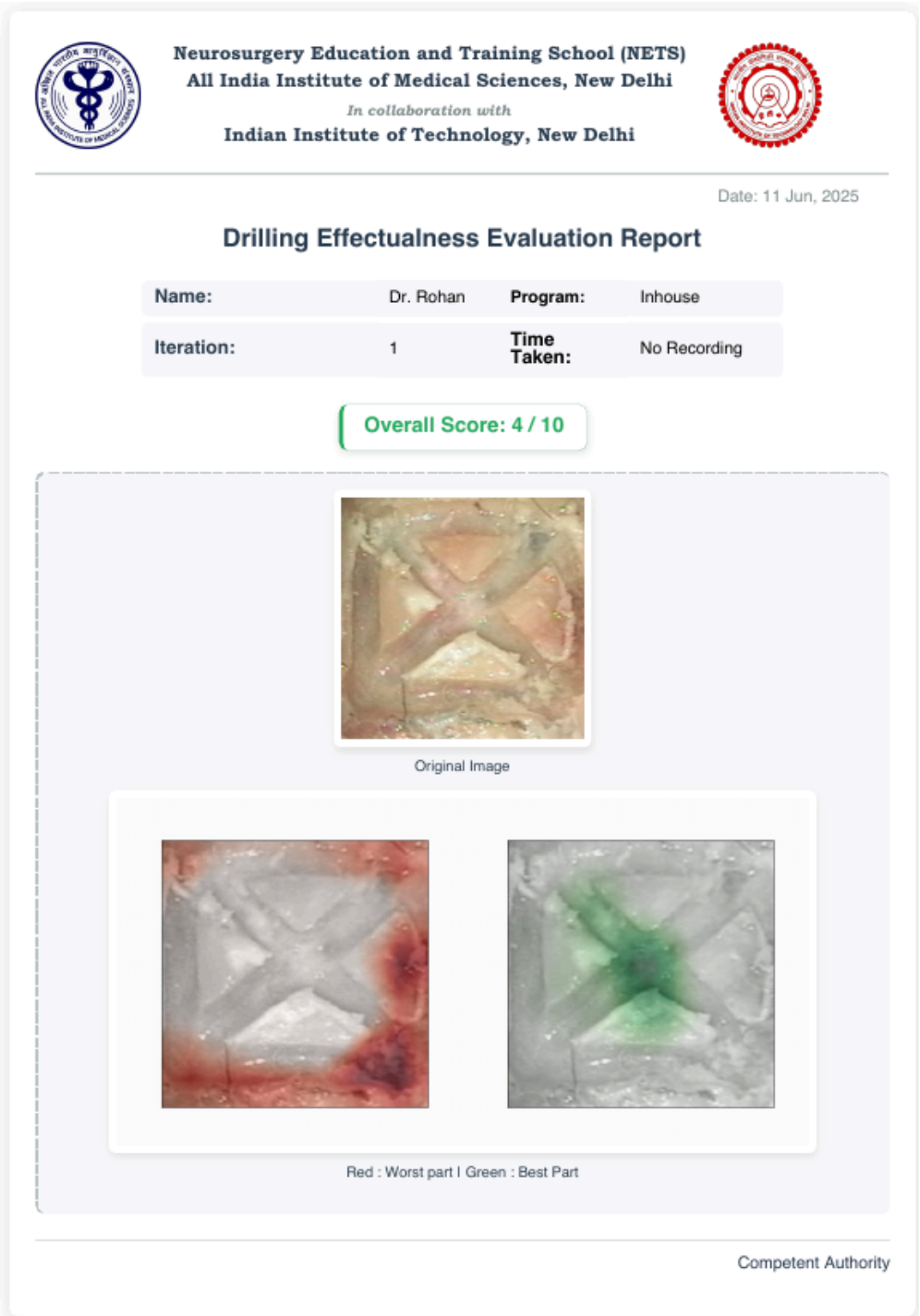
Below is the explanation for all the options provided in both the tabs

- Select Photo : Option which opens up file manager to choose an already saved static image for evaluation in photo mode, after choosing photo, another widow pops up where 4 corner points of the drilling sample have to be selected to homographically transform the image which will ultimately make model consider only that area which is formed by joining all the 4 chosen corners.
- Exit : Closes the Graphical user interface
- Camera Selection (Besides Exit) : Opens a drop down on all the available camera options and anyone can be selected for the feed.
- Evaluate : When the user clicks on this the image selected or captured is evaluated by the model and the result is updated in place of a placeholder in the final score.
- Generate Report : This option opens another window where the user has to enter his name details like name, email, iteration, thread size and program.
- Satisfied : Clicking on this will save the image and the scores
- Unsatisfied : This too does the same it saves the scores and image but it requires the user to enter the score they feel is suitable for sample image
- Submit feedback : This option is for the users to express their opinion on the project and give their feedback. Here the feedback entered by the user will be sent as a mail to the official nets lab email ID.
- Grad-Cam : Gradient Camera is a popular technique used in explainable AI to understand what is the basis for the models' decisions or scores.The grad-cam indicates or marks region with green and red color where regions marked with green color are the regions with best performance with respect to the task and the red regions are the areas with worst performance.

Additionally there are three options in the camera mode at the top,

- Capture Photo : This options captures the feed for whichever camera that is selected by user
- Start Recording / Pause Recording : Option to start recording the feed and this option leads to a window to save the file with a filename at the desired location and once started the same option is toggled for the option of Pause recording.
- Stop Recording : This option stops the recording and the video is auto saved at the location entered at the beginning.

Results and Conclusion



Neurosurgery Education and Training School (NETS)
All India Institute of Medical Sciences, New Delhi
In collaboration with
Indian Institute of Technology, New Delhi

Date: 11 Jun, 2025

Drilling Effectualness Evaluation Report

Name:	Dr. Rohan	**Program:**	Inhouse
Iteration:	1	**Time Taken:**	No Recording

Overall Score: 4 / 10

Original Image

Red : Worst part I Green : Best Part

Competent Authority

This above image is the report that is sent to users to their provided email ID.The stand alone application was successfully developed and deployed throughout the drilling lab.

ENDOSCOPY EYE TRACKING

This project aims to study the bio signals of the neurosurgeons while they perform the endoscopic hand-eye coordination training in which signals like eye tracking and tool movement etc are monitored closely to understand the patterns in the signal across neurosurgeons with different experiences and also to see that among people who are really good whether it's pure intuition and muscle memory or motor skills that yields the results.

Problem statement

To develop an eye tracking system which records the feed from three different cameras where two are ESP32S3 boards and one is a webcam. The two ESP32S3's have to be configured and then set up to provide a video stream to the system they are connected to.

Methodology

Arduino IDE was used to communicate to ESP32S3's and dump the c code into them such that they could capture frames at the required FPS and send the data to the system. As this project requires good quality and better frames per second (FPS) wired connection was opted instead of wireless as that would produce lag and frame drops which makes it difficult to analyze and get any outcomes from the data.Python was used to develop a graphical user interface and OpenCV was used to take care of all the video inputs and to record their feeds.

Results and Conclusion

The feed recording system was successfully built and deployed in the endo lab. More work was expected to be done on this project but as the tenure of internship got over, we could limit ourselves only to this much.

CONCLUSION

My internship at the Neuro-Engineering Lab at AIIMS has been an incredible and rewarding experience. It was a remarkable opportunity to apply my skills in AI and software development to the fascinating and highly specialized world of neurosurgery.

During my time in AIIMS from May to September 2025, I had the chance to work on several impactful projects. I've developed and deployed two key applications: a stand-alone system for giving surgeons real-time feedback on their craniotomy training , and a multi-camera system to capture data for endoscopic eye-tracking research. We also explored an innovative approach using agentic AI to evaluate cylindrical sutures. That particular project was a valuable lesson, offering a real-world look into both the immense potential and the current limitations of using large AI models for such a specialized medical task.

More than just building up my technical skills, this experience has given me a deep appreciation for the role technology can play in advancing medical training.

I am grateful to all the people from the lab for their unwavering support as it was one of the best learning and personal experiences for me.